

MODULE 08

CI/CD with GitHub Actions

Day 2 · Core MVP Module

CI/CD with GitHub Actions



Build + Test



Publish



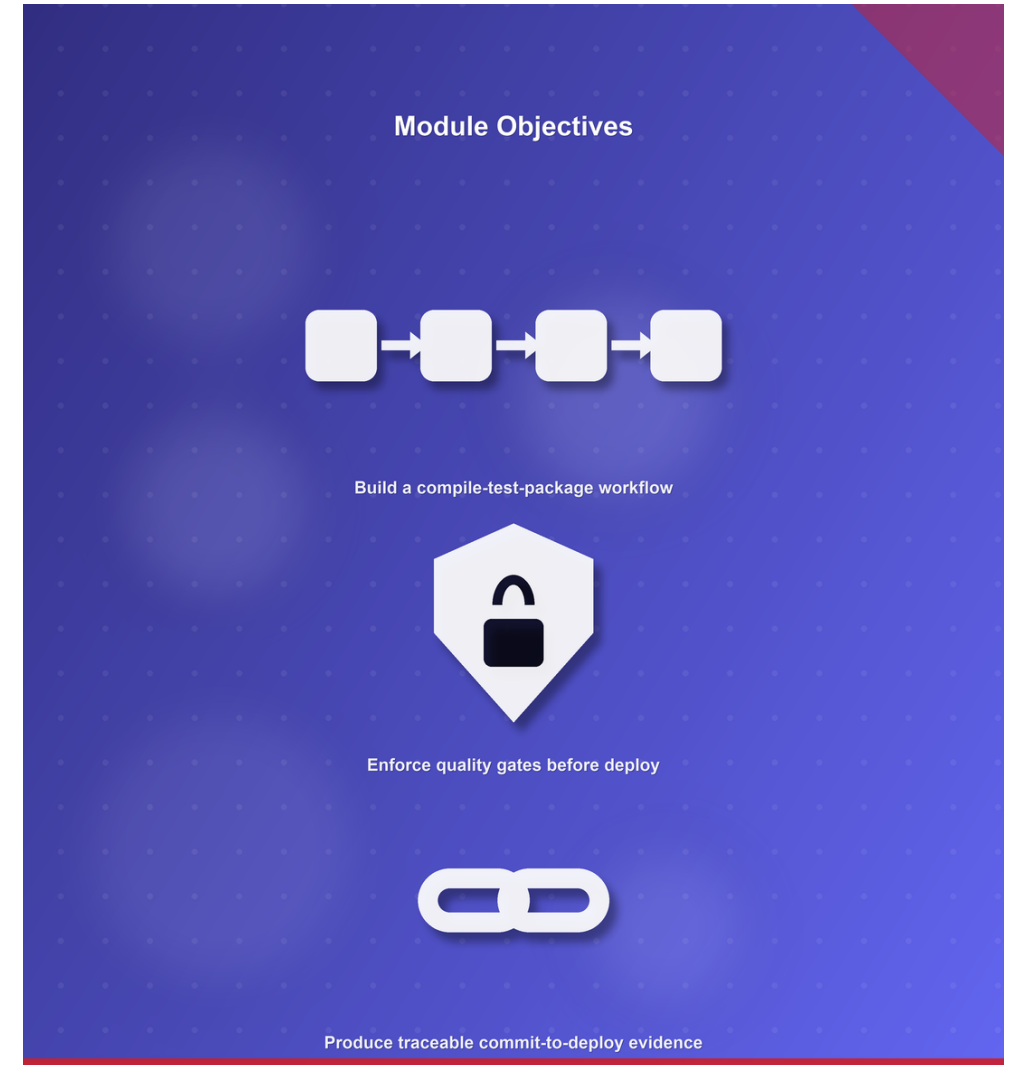
Deploy



Gate

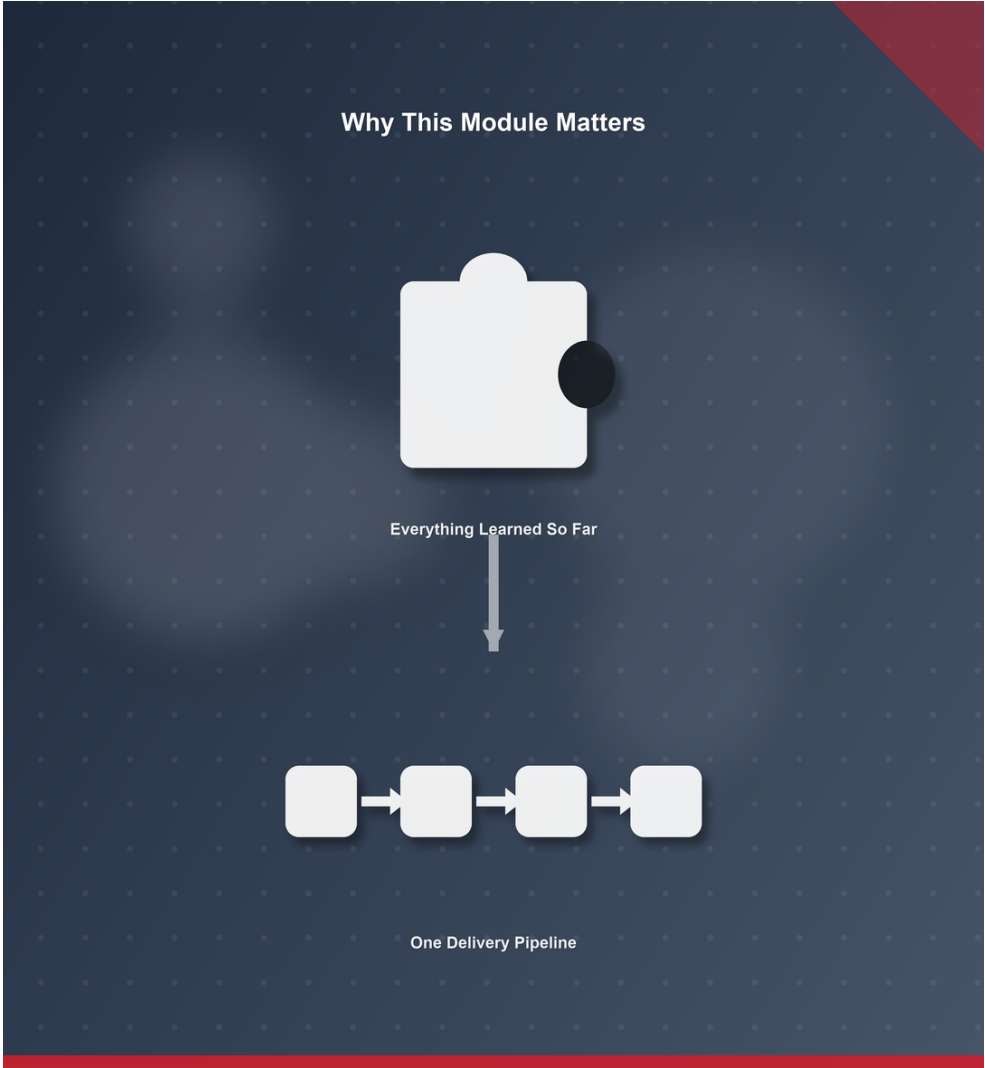
LEARNING OBJECTIVES

- Build a workflow that compiles, tests, and packages the application.
- Enforce quality gates before deployment.
- Produce traceable CI/CD evidence from commit to deploy.



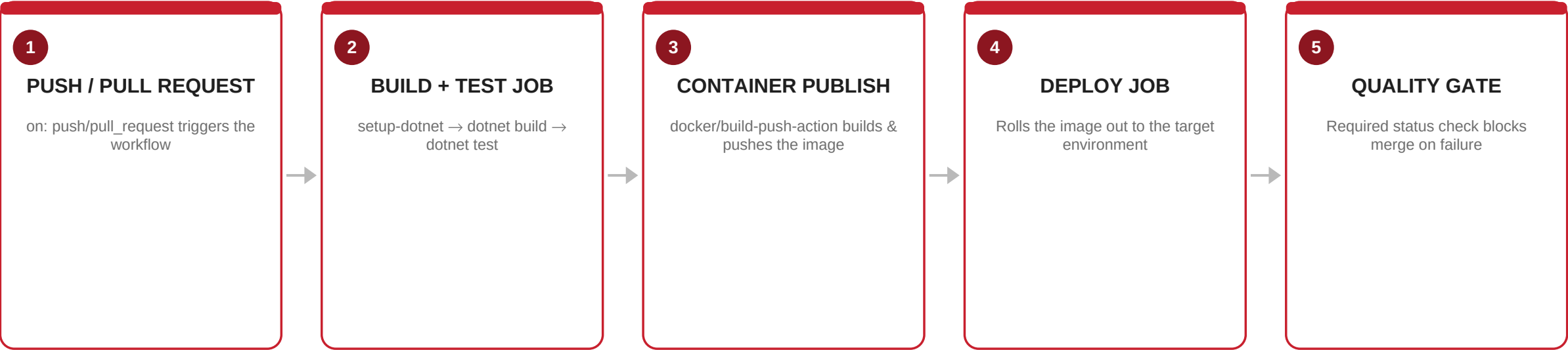
WHY THIS MODULE MATTERS

This module connects all prior modules into a delivery pipeline.
Automated checks and deployment orchestration tie the toolchain together.



CONCEPT MAP

Where eShopOnWeb's real workflow stops today, and the two stages Lab 08 adds.



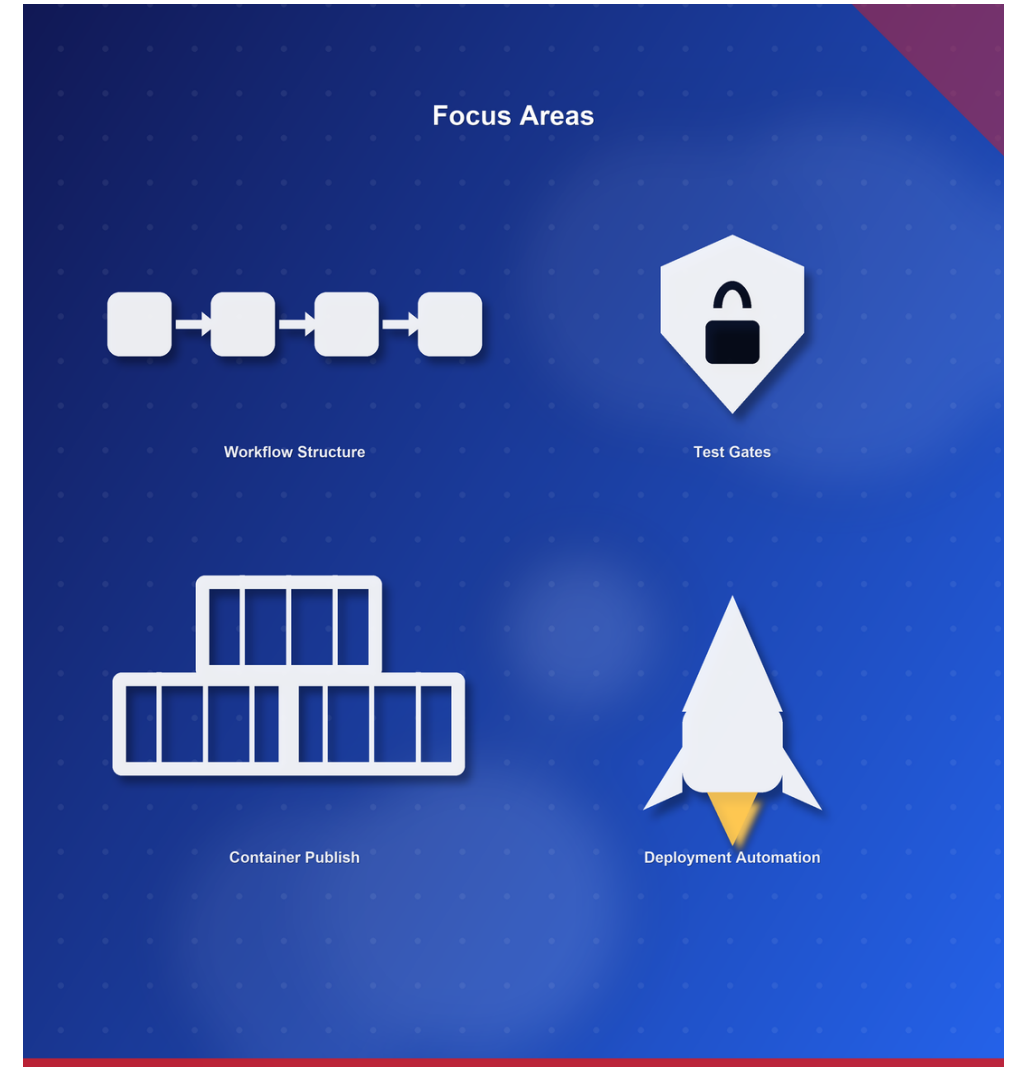
FROM THE DOCS: WHAT A QUALITY GATE ACTUALLY IS

- Workflows group work into jobs, and each job runs an ordered sequence of steps.
- actions/setup-dotnet finds a .NET version in the runner's tool cache and adds it to PATH.
- GitHub calls setup-dotnet the recommended way to get consistent .NET behavior across runners.
- A typical .NET CI job: dotnet restore, dotnet build --no-restore, dotnet test --no-build.
- Required status checks must pass before collaborators can merge into a protected branch.



COURSEWARE FOCUS

- Workflow structure
- Test gates
- Container publish
- Deployment automation



REAL WORKFLOW (TODAY) — ESHOPONWEB

REAL CODE — [.github/workflows/dotnetcore.yml](#)

```
name: eShopOnWeb Build and Test

on: [push, pull_request, workflow_dispatch]

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v2
      - name: Setup .NET
        uses: actions/setup-dotnet@v1
        with:
          dotnet-version: '8.0.x'
      - name: Build with dotnet
        run: dotnet build ./eShopOnWeb.sln --configuration Release
      - name: Test with dotnet
        run: dotnet test ./eShopOnWeb.sln --configuration Release
```

Build+test only, today. Lab 08 extends this exact file with a container-publish and deploy job.

TOOLS & RESOURCES

GitHub Actions

dotnet/setup-dotnet, docker/build-push-action, redhat-actions/oc-login

Modular pipeline steps for building, packaging, and deploying.

Lab Guide

Microsoft Learn: AZ-400 Implement CI/CD with GitHub Actions and Azure

Building end-to-end automation triggers and pipeline quality gates.

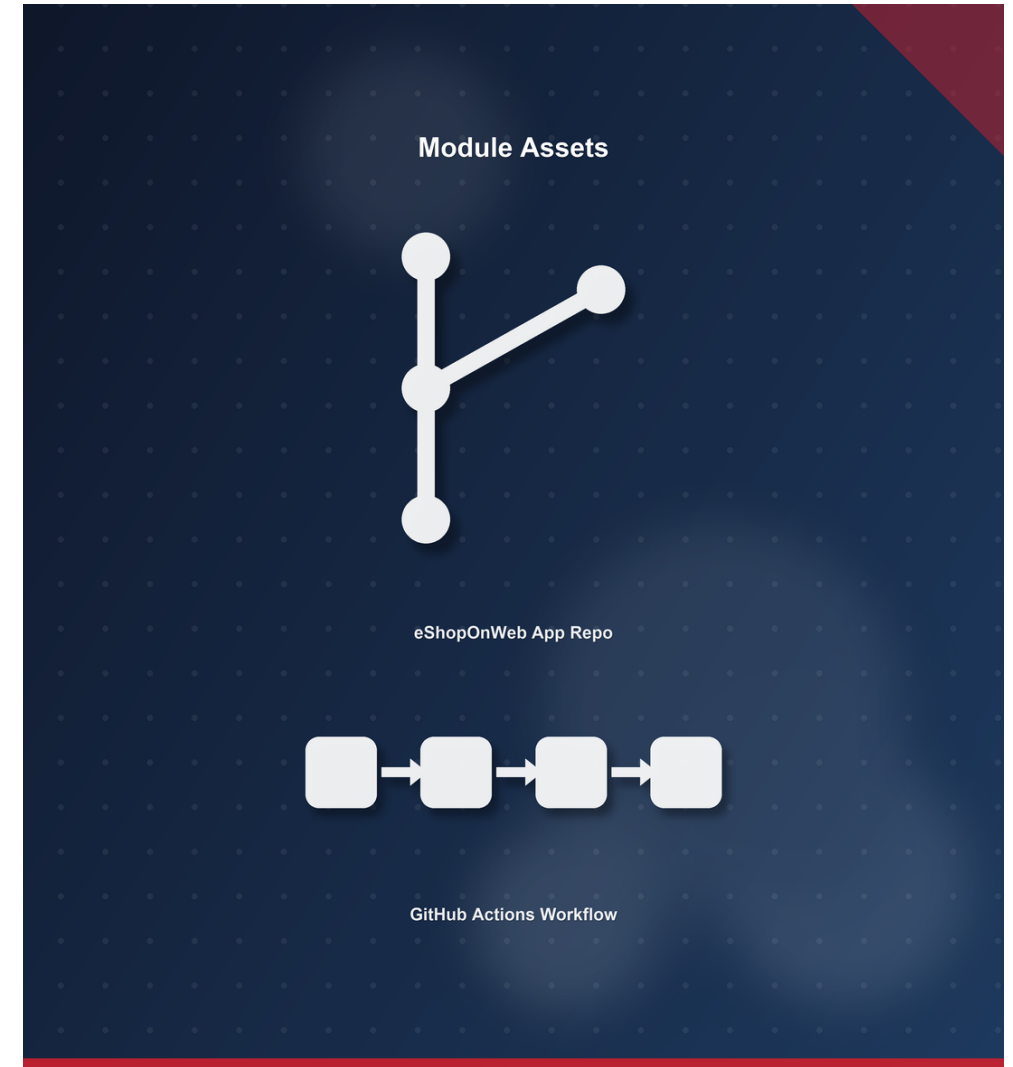
Sample Pipeline

.github/workflows/deploy.yml from the eShopOnWeb repo

Working reference workflow for container builds and deployment.

SUPPORTING ASSETS

- App repo from Modules 3 to 7
- Pipeline references from existing GitHub workflow patterns



LAB 08: CI/CD WITH GITHUB ACTIONS

Lab 08: CI/CD with GitHub Actions

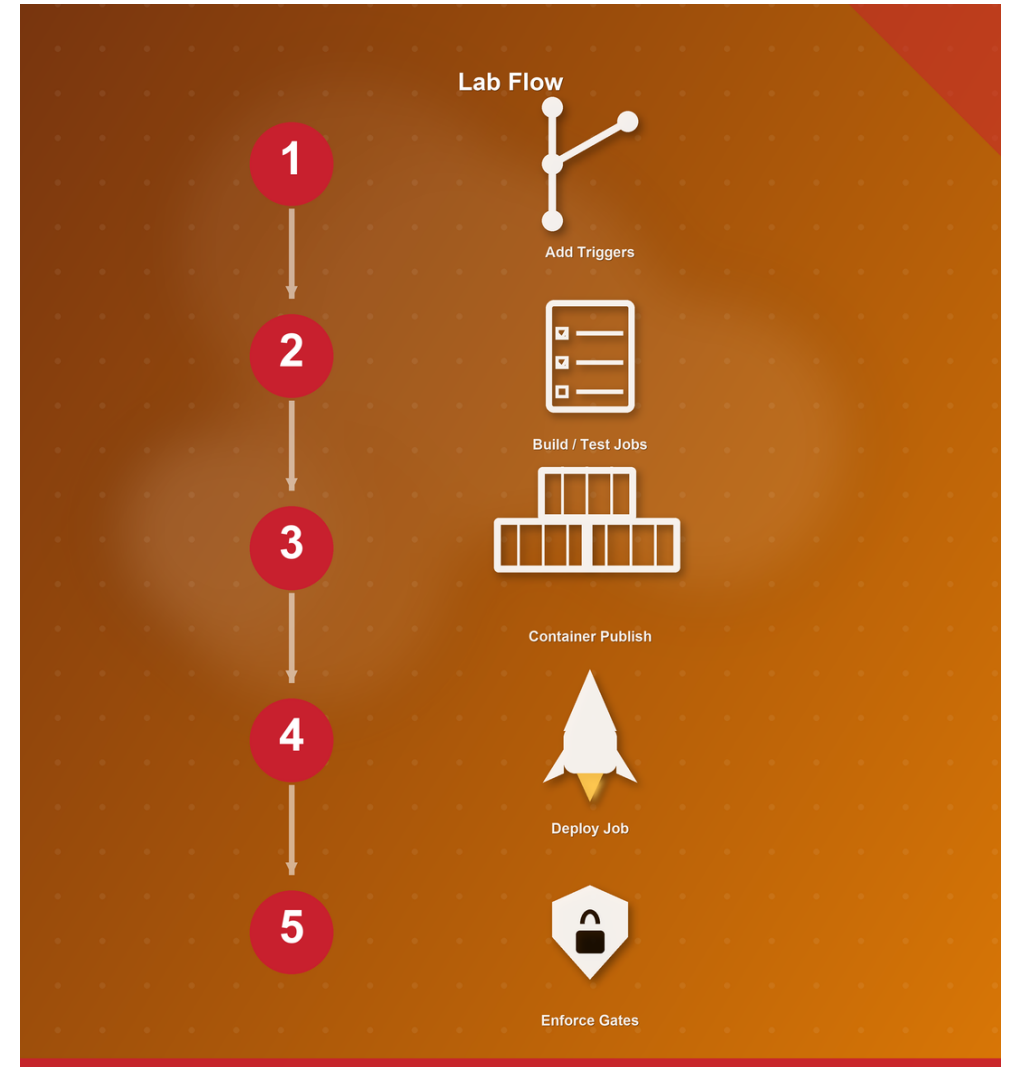


LAB STEPS

Goal: Create a working build-test-deploy workflow with quality gates.

Tools: Git, GitHub Actions · Requires GitHub secrets for registry/deploy targets.

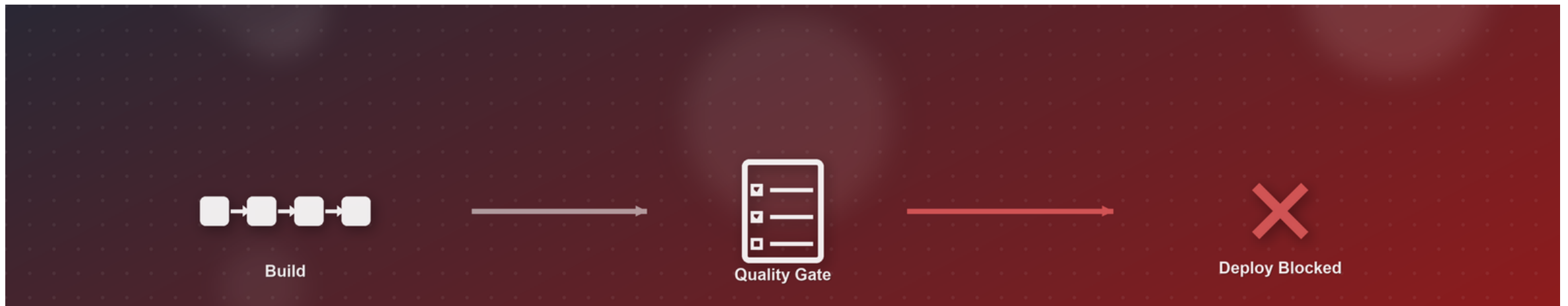
- Step 1 — Add workflow triggers for push and pull request.
- Step 2 — Add build and test jobs.
- Step 3 — Add a container build and publish stage.
- Step 4 — Add a deployment job and environment guardrails.
- Step 5 — Enforce test and coverage quality gates.
- Validation: the workflow executes end to end, and failed tests block the deploy stage.



KNOWLEDGE CHECK: CI/CD with GitHub Actions

In a CI/CD pipeline, what is a quality gate?

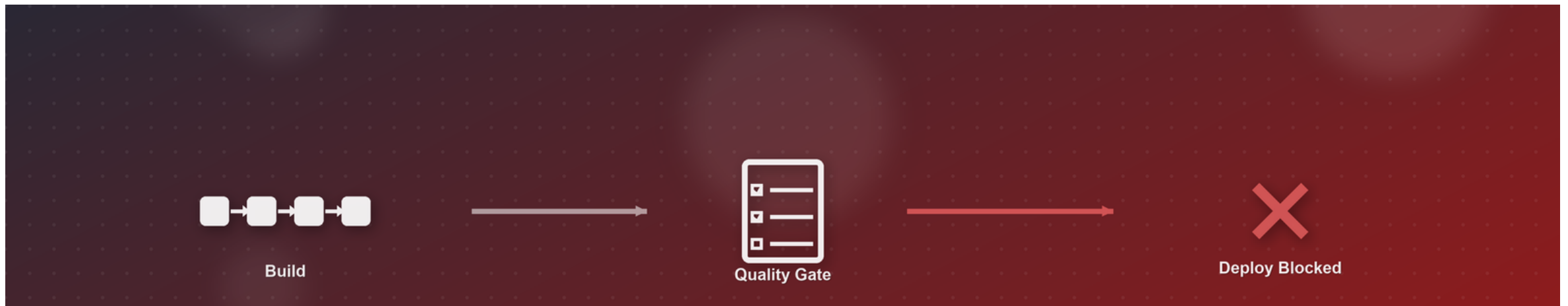
- A. A firewall rule
- B. A required check, such as passing tests, that must succeed before code proceeds to the next stage
- C. A type of Docker base image
- D. A GitHub repository naming setting



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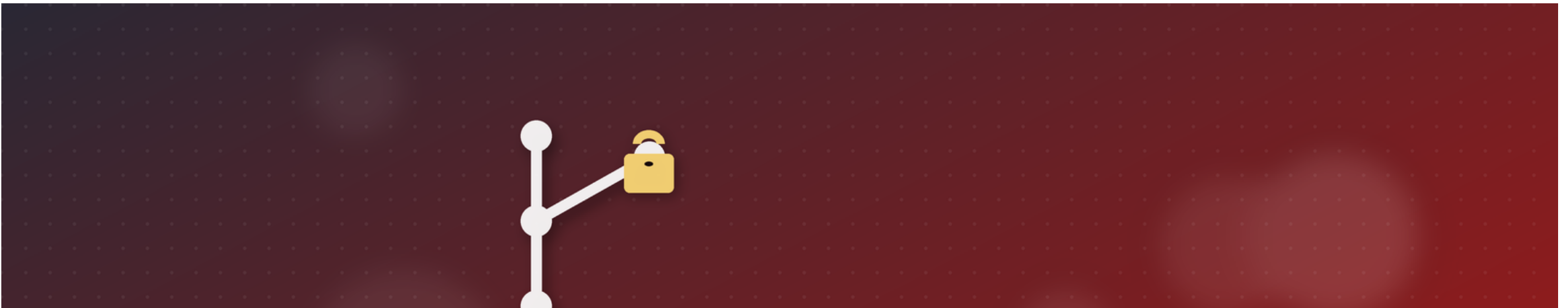
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KNOWLEDGE CHECK: CI/CD with GitHub Actions

Per GitHub's own docs, what must happen before collaborators can merge into a protected branch?

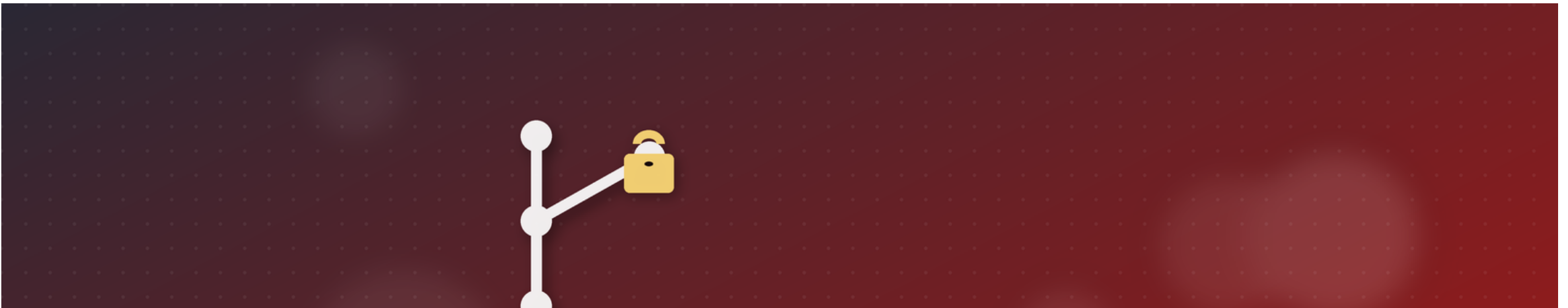
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- B. Required status checks must pass
- C. The repo must be archived
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SOURCES & FURTHER READING

EXTERNAL DOCUMENTATION

■ Building and testing .NET — GitHub Docs

<https://docs.github.com/actions/guides/building-and-testing-net>

REFERENCED IN THIS REPOSITORY

■ `course/repos/eShopOnWeb/.github/workflows/dotnetcore.yml`

■ `course/mvp-delivery/labs/lab-08-cicd-github-actions.md`

RECAP & SUCCESS CRITERIA

- Pipeline executes end to end and blocks failing quality checks.
- Tier: Core MVP Module

